20261112 Quiz

for (MATH102) Calculus II

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Problem 6.4. Show that the ball

$$B = \{\mathbf{X} \in \mathbb{R}^3 \mid \|\mathbf{X}\| \leq 1\}$$

is a smoothly bounded set.

Solution. The set B is closed and bounded, and its boundary is the unit sphere

$$\partial B = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}.$$

Put

$$D = \{(u, v) \in \mathbb{R}^2 \mid u^2 + v^2 \leq \tfrac{2}{3}\}.$$

This disk is smoothly bounded in $\mathbb{R}^2$, since its boundary circle is a finite union of C^1 coordinate graphs. On D, the functions

$$(u, v) \longmapsto \pm\sqrt{1 - u^2 - v^2}$$

are continuously differentiable: indeed, $1 - u^2 - v^2 \geq 1/3 > 0$ on D.

Using these functions, form the six graphs

$$\begin{aligned} x &= \pm\sqrt{1 - y^2 - z^2}, \quad (y, z) \in D, \\ y &= \pm\sqrt{1 - x^2 - z^2}, \quad (x, z) \in D, \\ z &= \pm\sqrt{1 - x^2 - y^2}, \quad (x, y) \in D. \end{aligned}$$

Each graph lies in ∂B. Conversely, if $(x, y, z) \in \partial B$, then

$$\max\{|x|, |y|, |z|\} \geq \frac{1}{\sqrt{3}},$$

because $x^2 + y^2 + z^2 = 1$. Thus at least one coordinate can be written as one of the six graphs above, with the other two coordinates belonging to D. Hence these six C^1 graphs cover ∂B.

Therefore B is smoothly bounded by Definition 6.4.

□

Problem 6.18. Write a Riemann sum for

$$\iint_{[0,1]\times[0,1]} x^2 y^3 \, \mathrm{d}A$$

using the upper right corner of each $\frac{1}{4}$-square as its sample point.

Solution. Divide $[0,1]\times[0,1]$ into the 16 squares

$$Q_{ij}=\left[\frac{i-1}{4},\frac{i}{4}\right]\times\left[\frac{j-1}{4},\frac{j}{4}\right],\qquad 1\le i,j\le 4.$$

Each square has area $\Delta A=1/16$, and its upper right corner is $(i/4,j/4)$. Thus the requested Riemann sum is

$$\boxed{\sum_{i=1}^{4}\sum_{j=1}^{4}\left(\frac{i}{4}\right)^{2}\left(\frac{j}{4}\right)^{3}\frac{1}{16}}.$$

□

Problem 6.20. Let $D=[a,b]\times[c,d]$, and let f and g be continuous on $[a,b]$ and $[c,d]$, respectively. Use Riemann sums to prove that

$$\iint_D f(x)g(y)\,\mathrm{d}A=\left(\int_a^b f(x)\,\mathrm{d}x\right)\left(\int_c^d g(y)\,\mathrm{d}y\right).$$

Solution. Let $h=2^{-n}$, as in the Riemann-sum construction of Section 6.2, and set

$$I_h=\{i\in\mathbb{Z}\mid [ih,(i+1)h]\subseteq[a,b]\},$$
$$J_h=\{j\in\mathbb{Z}\mid [jh,(j+1)h]\subseteq[c,d]\}.$$

The h-squares contained in D are exactly the products of the intervals indexed by I_h and J_h. Choose $x_i\in[ih,(i+1)h]$ for $i\in I_h$ and $y_j\in[jh,(j+1)h]$ for $j\in J_h$. The corresponding Riemann sum separates:

$$\begin{aligned}S(fg,D,h)&=\sum_{i\in I_h}\sum_{j\in J_h}f(x_i)g(y_j)h^2\\&=\left(\sum_{i\in I_h}f(x_i)h\right)\left(\sum_{j\in J_h}g(y_j)h\right).\end{aligned}$$

The intervals omitted at the ends of $[a,b]$ and $[c,d]$ have total lengths at most $2h$ in each case. Since f and g are continuous and therefore bounded, these omissions do not affect the limits of the one-variable Riemann sums. Consequently,

$$\lim_{h\to0}\sum_{i\in I_h}f(x_i)h=\int_a^b f(x)\,\mathrm{d}x,\qquad\lim_{h\to0}\sum_{j\in J_h}g(y_j)h=\int_c^d g(y)\,\mathrm{d}y.$$

Also, $f(x)g(y)$ is continuous on the rectangle D, so by the Riemann-sum convergence result of Section 6.2 (including its extension to functions of arbitrary sign), $S(fg,D,h)$ converges to $\iint_D f(x)g(y)\,\mathrm{d}A$. Taking the limit in the factorization gives

$$\boxed{\iint_D f(x)g(y)\,\mathrm{d}A=\left(\int_a^b f(x)\,\mathrm{d}x\right)\left(\int_c^d g(y)\,\mathrm{d}y\right)}.$$

□